

Vocabulary: Causal Inference and ML — Structural Causal Models and Deep Discovery

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Foundations of Causal Frameworks

- **Pearl’s Causal Hierarchy:** a tripartite categorization of information levels determining what types of reasoning are possible.

Level 1 is *Association* ($P(y | x)$, passive observation);

Level 2 is *Intervention* ($P(y | \text{do}(x))$, active manipulation);

Level 3 is *Counterfactuals* ($P(y_x | x', y')$, retrospective reasoning).

The fundamental constraint: traditional machine learning architectures are structurally confined to Level 1.

- **Structural Causal Model (SCM):** a system $\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F})$ mapping functional laws between observed and hidden variables. Formally:

$$X_j := f_j(\text{PA}_j, U_j), \quad j = 1, \dots, d$$

where PA_j represents immediate parent causes and assignments are inherently deterministic. *Unlike standard statistical equations, assignments are directional and cannot be algebraically inverted.*

- **Endogenous Variables ($\mathcal{V}$):** variables whose values are determined explicitly within the boundaries of the system by the structural functions $f \in \mathcal{F}$. *These constitute the nodes over which we perform empirical data collection and graph modeling.*
- **Exogenous Noise Variables ($\mathcal{U}$):** unobserved, background drivers originating outside the model boundaries that represent missing parameters or localized randomness. They are assumed to be jointly independent. *Exogenous noise allows an SCM to generate a valid, non-deterministic joint probability distribution over $\mathcal{V}$.*
- **Independence of Cause and Mechanism (ICM):** the axiom stating that the distribution of a cause $P(C)$ and the conditional mechanism mapping cause to effect $P(E | C)$ are completely autonomous modules. Changing $P(C)$ leaves $P(E | C)$ invariant. *The core reason why causal models generalize across distribution shifts where associative models fail.*
- **Structural Minimality:** the principle requiring that every parent variable listed in PA_j must possess a non-zero functional dependency on X_j . If an edge coefficient or function evaluates to zero, it must be removed. *A formal application of Occam’s Razor to graph structures, ensuring we avoid spurious edges.*

Structural Errors and Selection Bias

- **Confounder:** a variable that holds direct causal arrows pointing simultaneously into both the treatment (cause) and the outcome (effect). *Confounders create a backdoor path that generates spurious statistical correlations at Level 1.*
- **Simpson’s Paradox:** a mathematical contradiction where a statistical trend appears within distinct subgroups but completely reverses or disappears when the data is pooled together. *Caused by asymmetric treatment assignment weights across confounding subgroups; resolved cleanly via backdoor adjustment.*
- **Backdoor Adjustment Formula:** a technique to compute the true interventional effect by weighting conditional subgroup outcomes against their baseline population prevalence:

$$P(Y \mid \text{do}(X = x)) = \sum_z P(Y \mid X = x, Z = z)P(Z = z).$$

Effectively simulates a randomized controlled trial by mathematically slicing away the confounding influence of Z .

- **Collider Trap / Selection Bias:** a mistake that occurs when an engineer conditions on a common effect ($X \rightarrow Z \leftarrow Y$). Restricting data or adding Z as a feature forces an artificial, non-causal dependency between X and Y . *Example: Conditioning on celebrity status creates a false negative correlation between talent and appearance.*
- **Overcontrolling Bias:** a distortion created by conditioning on a mediator variable sitting along the path between the cause and effect ($X \rightarrow Z \rightarrow Y$). *Blocks the flow of the primary causal mechanism, making the target feature appear artificially weak or non-significant.*

Causal Discovery Assumptions

- **Causal Discovery:** the reverse-engineering task of learning the true Directed Acyclic Graph (DAG) or structural system directly from observational datasets.
- **Causal Markov Condition:** the foundational assumption stating that every node in a DAG is statistically independent of its non-descendants when conditioning on its immediate parents (PA_j). *Enables the factorization of the joint distribution into isolated conditional distributions.*
- **Faithfulness:** the structural assumption that all conditional independencies observed in a dataset are strictly a consequence of the DAG structure, rather than accidental parameter cancellations. *Rules out unfaithful systems where opposing causal paths happen to cancel each other out to zero.*
- **Causal Sufficiency:** the assumption that all shared, common confounders affecting any pair of variables in the system are fully observed and included in the dataset. *While essential for standard constraint-based algorithms, it is routinely violated in complex real-world data.*

Constraint-Based Discovery Equivalence

- **Markov Equivalence Class (MEC):** a collection of distinct DAGs that satisfy the exact same set of conditional independencies, making them statistically indistinguishable from passive observation alone. *For example, $A \rightarrow B$ and $A \leftarrow B$ belong to the exact same class.*

- **CPDAG** (Completed Partially Directed Acyclic Graph): a graph that visually represents a Markov Equivalence Class. It contains directed edges for invariant directions and undirected edges for reversible links.
- **Separating Set** $S(A, B)$: the collection of nodes that, when conditioned on, render variables A and B statistically independent ($A \perp\!\!\!\perp B \mid S$). *Tracking separating sets during edge pruning is critical for orienting downstream v-structures.*
- **PC Algorithm**: a classic constraint-based structure learning algorithm that uses conditional independence tests to isolate the skeleton of a graph, identify v-structures, and propagate remaining directions.
- **Unshielded Triple**: a trio of nodes $A - C - B$ where A is connected to C and B is connected to C , but A and B share no direct edge between themselves. *These configurations serve as the primary candidates for resolving directionality via collider checks.*
- **V-Structure / Collider Orientation**: the rule stating that if an unshielded triple $A - C - B$ drops its edge via a separating set $S(A, B)$ such that $C \notin S(A, B)$, the edges must be oriented into the middle node: $A \rightarrow C \leftarrow B$.
- **Meek Rules**: a set of four logically sound rules utilized to propagate orientations across a skeleton after v-structures are established, ensuring no new colliders or directed cycles are accidentally introduced.

Functional Form Adjustments

- **Additive Noise Model (ANM)**: a functional restriction where the structural assignment isolates noise to an additive component:

$$Y := f(X) + U_Y, \quad U_Y \perp\!\!\!\perp X.$$

If f is non-linear, this functional constraint breaks Markov equivalence, making two-node directionality fully identifiable.

- **Structural Residuals** $\hat{U}_Y$: the error values calculated by running a non-linear regression in a hypothesized causal direction ($\hat{U}_Y = Y - \hat{f}(X)$). *By checking whether $\hat{U}_Y \perp\!\!\!\perp X$ via independence tests, models can determine the true cause-and-effect direction.*

Common Shortnames

Shortname	Meaning
SCM	Structural Causal Model
DAG	Directed Acyclic Graph
MEC	Markov Equivalence Class
CPDAG	Completed Partially Directed Acyclic Graph
ICM	Independence of Cause and Mechanism
ANM	Additive Noise Model
$\text{do}(X)$	The do-operator representing an active intervention on variable X
PA_j	The immediate set of parents pointing directly into node j
$S(A, B)$	The separating set of variables that renders A and B independent
U_j	Exogenous noise term associated with structural assignment j